

Practical No- 4

Aim :- To develop a Customer Order Management System & retrieve data using SQL joins (INNER, LEFT, RIGHT, FULL OUTER) to understand relationships between multiple tables.

Theory :-

In database management system, joins are used to combine rows from two or more tables based on related column between them.

- 1) INNER JOIN :- Returns only the rows that have matching values in both tables.
- 2) LEFT JOIN (or LEFT OUTER JOIN) :- Returns all rows from the left table & matched rows from the right table. If there is no match, the result is NULL on the right side.
- 3) RIGHT JOIN (or RIGHT OUTER JOIN) :- Returns all rows from the right table & matched rows from the left table. If there is no match, the result is NULL on the left side.
- 4) FULL OUTER JOIN :- Returns all rows when there is a match in one of the tables. Unmatched rows from either table appear with NULL in the columns of the other table.

Tables Used :-

- 1) Customers :- Contains customer details.
- 2) Orders :- Contains order details linked to customers via Customer ID.

```
CREATE DATABASE IF NOT EXISTS StudentDB;
USE StudentDB;

SET FOREIGN_KEY_CHECKS = 0;
DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Customers;
SET FOREIGN_KEY_CHECKS = 1;

CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(50),
    ContactNumber VARCHAR(15)
);

CREATE TABLE Orders (
    OrderID INT PRIMARY KEY,
    CustomerID INT,
    OrderDate DATE,
    Amount DECIMAL(10,2),
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

INSERT INTO Customers (CustomerID, CustomerName, ContactNumber) VALUES
(1, 'Alice', '1234567890'),
(2, 'Bob', '2345678901'),
(3, 'Charlie', '3456789012'),
(4, 'David', '4567890123');

INSERT INTO Orders (OrderID, CustomerID, OrderDate, Amount) VALUES
(101, 1, '2026-03-15', 250.00),
(102, 2, '2026-03-16', 450.00),
```

```
(103, 2, '2026-03-17', 150.00),  
(104, 3, '2026-03-18', 300.00);  
  
SELECT c.CustomerID, c.CustomerName, o.OrderID, o.Amount  
FROM Customers c  
INNER JOIN Orders o  
ON c.CustomerID = o.CustomerID;  
  
SELECT c.CustomerID, c.CustomerName, o.OrderID, o.Amount  
FROM Customers c  
LEFT JOIN Orders o  
ON c.CustomerID = o.CustomerID;  
  
SELECT c.CustomerID, c.CustomerName, o.OrderID, o.Amount  
FROM Customers c  
RIGHT JOIN Orders o  
ON c.CustomerID = o.CustomerID;  
  
SELECT c.CustomerID, c.CustomerName, o.OrderID, o.Amount  
FROM Customers c  
LEFT JOIN Orders o  
ON c.CustomerID = o.CustomerID  
  
UNION  
  
SELECT c.CustomerID, c.CustomerName, o.OrderID, o.Amount  
FROM Customers c  
RIGHT JOIN Orders o  
ON c.CustomerID = o.CustomerID;
```

Result Grid



Filter Rows:

Export:



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	CustomerID	CustomerName	OrderID	Amount
▶	1	Alice	101	250.00
	2	Bob	103	150.00
	2	Bob	102	450.00
	3	Charlie	104	300.00
	4	David	NULL	NULL

Conclusion :-

- 1) INNER JOIN :- Showed only Customer who have placed orders.
- 2) LEFT JOIN :- displayed all Customers, including those without orders.
- 3) RIGHT JOIN :- displayed all orders, even those placed by customers not present in Customers table.
- 4) FULL OUTER JOIN :- provided a complete view of both tables, including unmatched rows.